

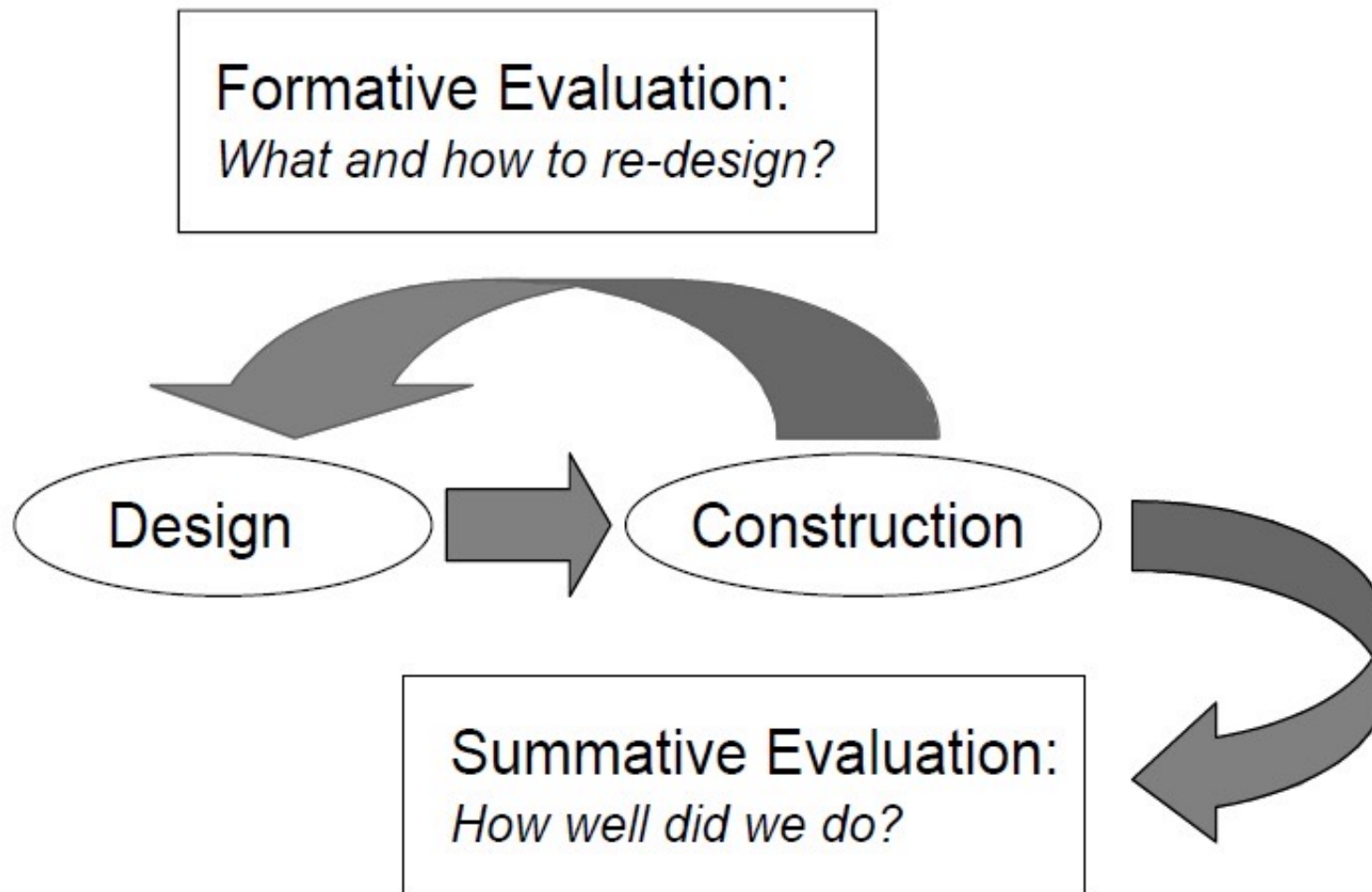
Usability Evaluation – Part I

Corso di Informatica – Laurea Magistrale

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Goals of Usability Evaluation



Formative and Summative Goals

Formative: *during development, guides process*

Summative: *after development, or at a checkpoint*

- What sorts of “test data” aid formative goals?
- What about summative?
- SBD relies on mediated usability evaluation
 - claims analysis documents design features of concern
 - users’ performance & reactions tied to these features, establishing a *usability specification*

Analytic and Empirical Methods

Analytic: theory, modeling, guidelines (from *experts*)

Empirical: observations, surveys (from *users*)

"If you want to evaluate a tool, say an axe, you might study the design of the bit, the weight distribution, the steel alloy used, the grade of hickory in the handle, etc., or you might just study the kind and speed of the cuts it makes in the hands of a good axeman."

ANALYTICAL

- Which is more expensive? Why?
- Which carries more weight with developers? Why?

EMPIRICAL

Usability Inspection

- Expert walk-through based on usability guidelines, often working from a checklist
 - generally want more than one expert (if affordable!)
- Guidelines (and walkthrough) can be at many levels
 - e.g., screen layout, detailed analysis of cognitive states
 - (e.g., you may use Nielsen's guidelines)
- May or may not use a standard set of tasks
 - depends on how comparable you want judgments to be
- Summarize by listing problems identified in each category, also often rating them for *severity*

Heuristic Evaluation

Nielsen's general guidelines

- Use simple and natural language
- Speak the users' language
- Minimize memory load
- Be consistent
- Provide feedback
- Provide clearly marked exits
- Provide shortcuts
- Provide good error messages
- Prevent errors
- Include good help and documentation

Multiple experts review with respect to these issues

Can also include different classes of stakeholders, e.g. developers, users

Cognitive Walkthrough

- Form-based inspection of system or prototype
 - for walk up and use systems (affordances, metaphors)
 - careful task selection, then answer questions at each step
 - e.g., how obvious is the next action? what knowledge is assumed? what if any competing goals must be ignored?
- Check-list approach is attractive to practitioners
 - concerns with how to select the tasks for analysis, i.e. complexity/realism trades off with cost of evaluation
 - in practice, often can be used in more lightweight fashion, more of a “tuning” to issues

Cognitive Walkthrough (cont.)

To perform a cognitive walkthrough you need to have:

1. Description of the prototype of the system
2. Description of the representative tasks that the user performs on the system
3. Complete list of actions needed to complete the task using the prototype
4. Indications of who the users of the product are and their type of experience and knowledge

Cognitive Walkthrough (cont.)

- The evaluators follow step by step the sequence of actions described in point 3
- For each action they will have to answer the following questions:
 - ❖ Will the correct action be obvious enough to the user? (Will the user know what to do to accomplish the task?)
 - ❖ Will the user notice that the correct action is available? (Can users see the button or menu item they should use for the next action? Is it obvious when it's needed?)
 - ❖ Will the user match and interpret the action response correctly? (Will users know from the feedback that they made a correct or wrong action choice?)

Cognitive Walkthrough Form

Evaluator: _____

Date: _____

Task: _____

1. Correct Action Obvious?

Will the user know what to do to accomplish the task?

Step	Expected Action	Obvious? (Y/N)	Notes / Confusion

2. Action Noticeable?

Can users see the button/menu item they need and recognize when it's needed?

Step	UI Element / Control	Visible / Discoverable? (Y/N)	Notes / Suggestions

3. Feedback Interpreted Correctly?

Will users understand the system's response to their actions?

Step	User Action	System Feedback	Clear Success/Failure? (Y/N)	Notes / Recommendations

GOMS Analysis

- Build *predictive model using scientific knowledge* about human memory and behavior
 - like HTA, can analyze for complexity, consistency
 - or build computational version, to estimate task times for different design alternatives
 - if successful, can provide huge benefit...why?
- Extends general techniques of HTA
 - goals, subgoals, plans, actions
 - BUT adds model elements for *mental activities such as* goal creation and selection, memory retrieval, etc

a GOMS snippet

```
GOAL:  CLOSE-ACTIVE-WINDOW
├─ [select GOAL:  USE-MENU-METHOD
    │   └─ MOVE-MOUSE-TO-MENU-BAR
    │       DRAG-DOWN-FILEMENU
    │       RELEASE-ON-CLOSE-OPTION
    └─ GOAL:  USE-HANDLE-METHOD
        └─ MOVE-MOUSE-TO-CORNER
            CLICK-ON-CLOSE-BOX
    └─ GOAL:  USE-CONTROL-KEY
        └─ PRESS-APPLE+W]
```

Downsides of Analytic Methods

- Usability inspections are rapid, relatively cheap
 - **but may miss details only seen in realistic use** contexts involving real users, say little about what caused the problems, or expected impact
- Model-based approaches have good scientific foundation, are credible, can be very powerful
 - **but current theories have limited scope, and** developing the models takes time/expertise

DECIDE: A Framework to guide Empirical Evaluation

- **Determine** the *objectives* that the evaluation intends to pursue.
- **Explore** the specific *questions* that need to be answered.
- **Choose** the *paradigm* and the *evaluation techniques* to answer the questions.
- **Identify** the *practical issues*.
- **Decide** how to handle the *ethical issues*.
- **Evaluate**, interpret, and present the *data*.

Determine the Objectives

- **What are the high-level goals of the evaluation?**
- **Who wants it, and why?**

The goals influence the paradigm chosen for the study.

Some examples of goals:

- Identifying the best metaphor on which to base the design.
- Checking that the final interface is coherent.
- Investigating how the technology influences work practices.
- Improving the usability of an existing product.

Explore the specific *questions*

- All evaluations need objectives and guiding questions so that time is not wasted on poorly designed studies.
- For instance, the goal of understanding why users frequently abandon an online shopping cart before completing checkout can be broken down into sub-problems:
 - Are users confused by the number of steps in the checkout process?
 - Do unexpected costs (e.g., shipping fees) cause them to quit?
 - Is the payment interface difficult to use or perceived as insecure?
 - Are there technical issues such as slow page loading or errors?

Choose the *paradigm* and the *evaluation techniques*

- **Which paradigm?**
 - *Quick and dirty* evaluation (rapid user feedback on specific issues)
 - Laboratory study
 - Field study
 - Heuristic evaluation

The evaluation paradigm strongly influences the techniques used and how the data are analyzed and presented.

For example, field studies do not involve testing or modelling.

Identify the *practical issues*

For example, how to:

- select the users
- stay within budget
- meet the deadlines
- find the evaluators
- select the equipment

Decide on *ethical issues*

- Develop an **informed consent form**
- Participants have the right to:
 - know the objectives of the study
 - know what will happen to the study results
 - have their personal data kept private
 - not be quoted without their consent
 - withdraw from the study at any time
 - be treated politely

Evaluate, interpret, and present the *data*

- How the data are analyzed and presented depends on the paradigm and the techniques used.
- You must also consider the following:
 - **Reliability:** can the study be replicated?
 - **Validity:** is it measuring what you intended to measure?
 - **Bias/Distortion:** is the process introducing any distortions?
 - **Scalability/Generalizability:** can the results be generalized?
 - **Ecological validity:** does the environment in which the study is conducted influence the results?

Empirical Evaluations: Validity

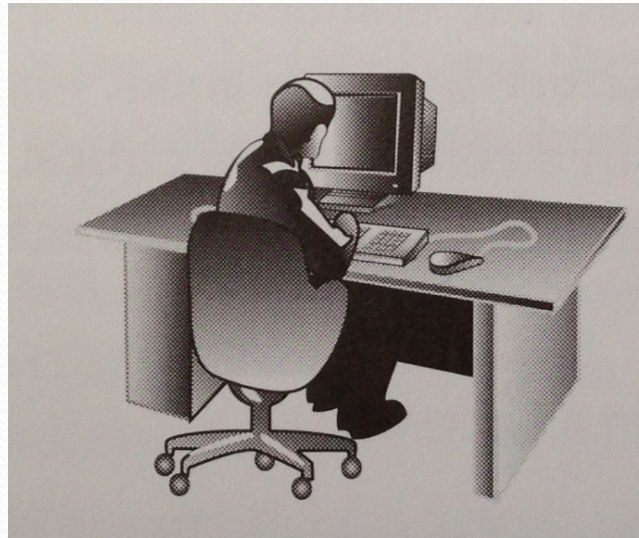
- Conclusions based on actual use, BUT...
 - are the users representative?
 - is the test population large, diverse enough?
 - is the test system realistic (versus early prototype)?
 - do the tasks match what happens in real use?
 - do the data (measures) reveal real life impacts?
- These are the general concerns of “ecological validity”, the extent to which an investigation is a genuine reflection of real-world happenings

Validity concerns

Is this really the kind of person who will use our system?

Is the prototype missing any important feature?

Will people be more distracted in their office?



How much of what I see is specific to this user?

Will our actual users do tasks like these?

Test participant working on a task in a usability lab



Field Studies

- Variants of ethnographic methods we discussed during requirements analysis
 - observation of realistic tasks, interviews, data files, etc.
 - avoids the problem of ecological validity
- Summarize data through content classification
 - e.g., problem categories, as in themes analysis
 - can also sort by severity, based on observed impacts
- Field data collection and analysis time-consuming
 - also, much of the data simply reveals successful use!

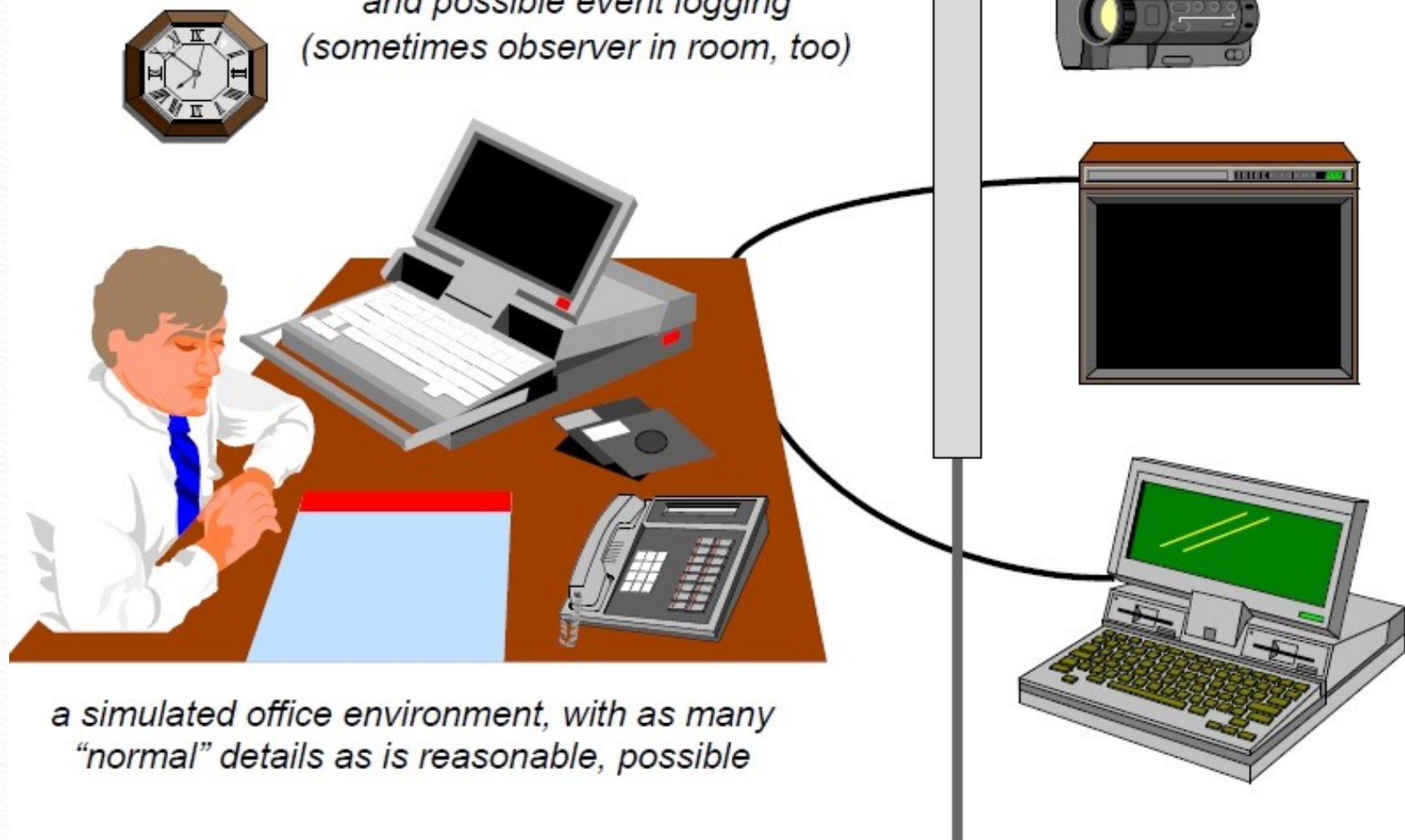
Interviews, User Introspection

- Ask users to report what stands out in memory
 - more efficient access to data of interest: *critical incidents*
 - can enhance by making this collaborative, a discussion among usability personnel and multiple stakeholders
- BUT, human memory is biased by many things
 - wanting things to make sense, assuming things work as they always have

Usability Evaluation in the Lab

- Carefully selected set of representative tasks
 - e.g., based on task analysis of the system, design goals
 - in SBD, **claims** are used to guide task selection
- Control aspects of situation that are uninteresting
 - e.g., experimenter, location, task order, instructions
- Collect multiple measures of usability impacts
 - performance (time and errors), output quality
 - satisfaction ratings or other subjective measures
- Interpretation comes back to validity of the test
 - both ecological (realism) and internal (controls)

observational data collected via a one-way mirror, video cameras, screen capture, and possible event logging (sometimes observer in room, too)



a simulated office environment, with as many "normal" details as is reasonable, possible

Usability Specifications

- Quality objectives for final system usability
 - like any specification, must be precise
 - managed in parallel with other design specifications
- In SBD, these come from scenarios & claims
 - scenarios are analyzed as series of critical subtasks
 - reflect issues raised and tracked through claims analysis
 - each subtask has one or more measurable outcomes
 - tested repeatedly in development to assess how *well* project is doing (summative) as well as to direct design effort toward *problem areas* (*formative*)
- Precise specification, but in a context of use

Activity, information, and interaction claims:

identify key design features to be tested

Design scenarios:

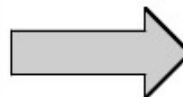
extract motivation and context for subtasks to be tested

Estimates of behavior:

published or pilot data of expected user behavior

Usability specifications:

a list of subtasks with performance and satisfaction parameters



Digital Library Search Subtask

Usability Outcomes Expected

Accessing the digital library

A user with at least one hour of previous usage experience should be able to access the main page of the digital library in 20 seconds or less, with no errors, and should rate "ease of access" no less than 4 on a 5-point rating scale.

Initiating the search function

A user with at least one hour of previous usage experience should be able to initiate the search dialog from the main page of the digital library in 5 seconds or less, with no errors, and should rate "search availability" no less than 4 on a 5-point rating scale.

Specifying search keys
for Scriven's article

A user with at least one hour of previous usage experience should be able to specify search terms that will successfully retrieve the Scriven reference in 30 seconds or less, with one error, and should rate "usefulness of search" no less than 4 on a 5-point rating scale.

What about Generality?

- Salient risk in focusing only on design scenarios
 - may optimize for these usage situations
 - the “successful” quality measures then reflect this
- When possible, add contrasting scenarios
 - overlapping subtasks, but different user situations (user category, background, motivation)
 - assess performance satisfaction across scenarios
- Motivation to construct functional prototypes as early as feasible in development cycle

Working from a VSF scenario

(excerpts from the teacher-coaching scenario...)

- when Mr. King meets Sally in the VSF, he can see she is already there, so he selects her name and uses Control+I to see that she is working on her slides, then Control+F to synchronize with her
- he watches her work, and sees her uploading files from her desktop using a familiar Windows browse-file dialog when he sees an Excel document, he experiments to see if it is 'live', discovers he can edit but not save
- when he sees that she is planning to have visitors come up with their own results using her simulation, he advises her that this will crowd the display, and goes off to find a way to create a 'nested' display element
- ...

Associated Claims

- Using “Control-I” to identify activities of co-present user
 - + ties info about people directly to their representation on the display
 - + simplifies the screen display by hiding activity information cues
 - but conflicts with the real world strategy of just looking around
 - but this special key combination must be learned
- Exhibit components shown as miniaturized windows
 - + suggests they may contain interactive content
 - but viewers may interpret them as independent applications
- File-browsing dialogs for uploading workstation documents
 - + builds on familiarity with conventional client-server applications
 - + emphasizes a view of exhibits as an integration of other work
 - but the status of these personal files within the VSF may be unclear

From Claims to Subtasks

- Identifying and joining co-present users
 - we know that key combinations are harder to learn; how distracting or difficult are they in this case?
- Recognizing and working with active components
 - will users understand these as 'active' objects? will they know how to activate them? will they know what is possible when they have done this?
- Importing desktop files into the VSF
 - is the operation intuitive, smooth? is there any resulting confusion about the status of the uploaded files?

A Sample Usability Specification

Scenario & Subtasks	Worst Case	Planned	Best Case
Interaction Scenario: Mr. King coaches Sally	2.5 on usefulness, ease of use, and satisfaction	4 on usefulness, ease of use, and satisfaction	5 on usefulness, ease of use, and satisfaction
1. Identify Sally's view and synchronize	1 minutes, 1 error 3 on confusion	30 seconds, 0 error 2 on confusion	10 seconds, 0 error 1 on confusion
2. Upload desktop file from the PC	3 minutes, 2 errors 3 on familiarity	1 minute, 1 error 4 on familiarity	30 seconds, 0 error 5 on familiarity
3. Open, modify, try to save Excel file	2 minutes, 1 error 3 on confidence	1 minute, 0 errors 4.5 on confidence	30 seconds, 0 error 5 on confidence
4. Create nested exhibit component	5 minutes, 3 errors 3 on complexity	1 minute, 1 error 2 on complexity	30 seconds, 0 error 1 on complexity

- Where do targets come from? They are serious, but not absolute

Esercizio sul Cognitive Walkthrough

- Considerate lo storyboard che illustra come un utente esegue il task «prenota tavolo» sull'app *EasyDinner*.
- Provate ad applicare la tecnica del cognitive walkthrough a tale task
 - Per ogni azione rispondete alle domande del cognitive walkthrough.
 - Commentate gli aspetti critici che risultano dal walkthrough

Storyboard: prenota da EasyDinner

